UNIT – II

Plane Motion Analysis

Subject : Theory of Machines (23A03403)

Syllabus: Displacement, velocity & acceleration analysis of simple mechanisms, graphical velocity analysis using instantaneous centres, analysis using loop-closure equations, kinematic analysis of simple mechanisms, slider-crank dynamics, coincident points, Coriolis component of acceleration.

1. Introduction – Motion of a Link

Plane motion of a rigid link = translation + rotation. Two points A and B of the same rigid link keep a constant distance, so the velocity of B relative to A is perpendicular to AB :

$$v_{BA} = \omega \times AB \ (\perp AB) \quad v_B = v_A + v_{BA}$$

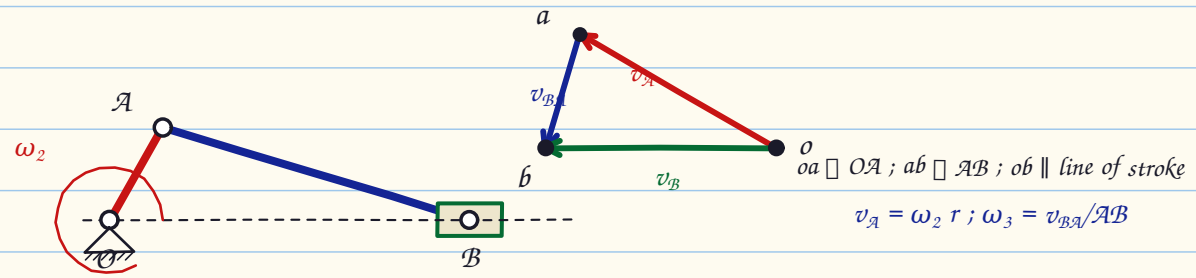
- **Acceleration** of B relative to A has two components : radial (centripetal) $a^r_{BA} = \omega^2 \, AB = v_{BA}^2/AB$ (directed from B toward A) and tangential $a^t_{BA} = \alpha \cdot AB$ ($\perp AB$).

$$a_B = a_A + a^r_{BA} + a^t_{BA}$$

- Angular velocity of a link = v_{BA}/AB (relative velocity of any two points ÷ distance) ; angular acceleration $\alpha = a^t_{BA}/AB$.
- A point on a rotating crank of radius r has $v = \omega r$, $a_n = \omega^2 r$ (toward the centre), $a_t = \alpha r$.
- **Methods of analysis** : (i) graphical – relative velocity (velocity polygon), instantaneous centre, Klein's construction ; (ii) analytical – loop-closure vector equations, complex numbers, matrices ; (iii) computer / numerical.

2. Displacement, Velocity & Acceleration of a Slider-Crank

Configuration diagram Velocity polygon (relative velocity)



Crank OA = r, connecting rod AB = l, n = l/r = obliquity ratio, crank angle θ measured from inner dead centre (IDC), φ = obliquity angle of connecting rod : sinφ = sinθ/n.

$$x = r\,(1 - \cos\theta) + l\,(1 - \cos\varphi) \approx r\,[\,(1 - \cos\theta) + \sin^2\theta/(2n)\,]$$ (piston displacement from IDC)

- Piston velocity : $v = r\,\omega\,(\sin\theta + \sin 2\theta/(2n)) = r\omega\,\sin(\theta+\varphi)/\cos\varphi$ (exact).
- Piston acceleration : $a = r\,\omega^2\,(\cos\theta + \cos 2\theta/n)$ ($a = d\,v/dt$, $\alpha_{crank} = 0$).
- Angular velocity of connecting rod $\omega_3 = \omega\,\cos\theta\,/\sqrt{(n^2 - \sin^2\theta)}$; ω_3 is max at θ = 0 and 180°, zero at the dead centres of the rod (θ = 90°, 270° for large n).
- Acceleration of the piston is maximum at the dead centres : $a = r\omega^2(1 + 1/n)$ at IDC (θ = 0) and $-r\omega^2(1 - 1/n)$ at ODC ; velocity is maximum at about θ = 75°–80° (n = 4 : 1.03 rω, slightly more than the crank-pin speed). For $n \to \infty$ the motion becomes SHM ($v = r\omega\,\sin\theta$, $a = r\omega^2\cos\theta$).

θ (deg)	x/r	v/(rω)	a/(rω²)	φ (deg)
0	0	0	+1.250	0
30	0.165	0.609	+0.991	7.2
60	0.595	0.977	+0.375	12.5
90	1.127	1.000	−0.250	14.5
120	1.595	0.755	−0.625	12.5
150	1.897	0.391	−0.741	7.2
180	2.000	0	−0.750	0

Table for n = l/r = 4 (θ from IDC). Note the piston covers more than half its stroke (x/r = 1.127 at 90°) in the first 90° of crank – asymmetry due to the finite rod

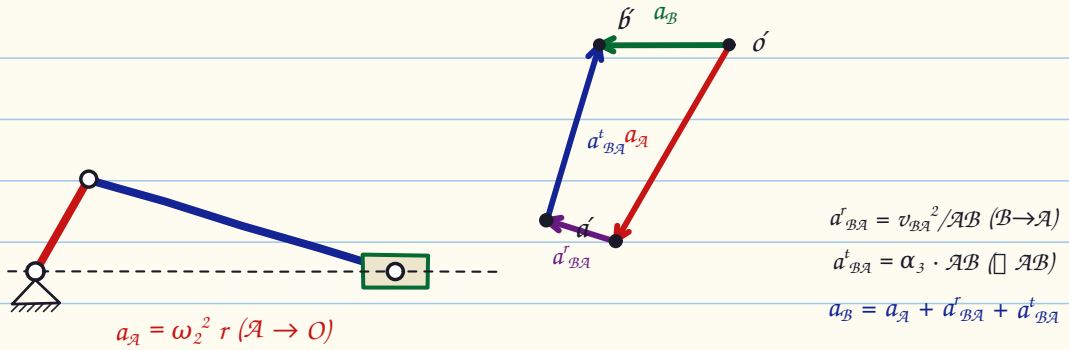
length.

> Ex 1. $r = 100$ mm, $l = 400$ mm ($n = 4$), $N = 300$ rpm ($\omega = 31.42$ rad/s), $\theta = 45°$: $v = 0.1 \times 31.42\,(0.7071 + 1/8) = 2.61$ m/s (exact 2.62) ; $a = 0.1(31.42)^2(0.7071 + 0) = 69.8$ m/s² ; $\omega_3 = 31.42 \times 0.7071/\sqrt{(16 - 0.5)} = 5.64$ rad/s.

Relative velocity (velocity polygon) method – steps

- (1) Draw the configuration diagram to scale. (2) $v_A = \omega_2\, r \perp OA$ – draw oa. (3) From a draw ab $\perp$ AB (direction of v_{BA}) ; from o draw ob along the line of stroke ; intersection = b. (4) v_B = ob, v_{BA} = ab, ω_3 = ab/AB. (5) The velocity of any point C on AB : locate c on ab such that ac/ab = AC/AB (velocity image).
- Angular velocity sense : v_{BA} direction relative to A shows rotation of AB (CW/CCW).

$\alpha_2 = 0$ (uniform crank speed) Acceleration polygon



Acceleration polygon (graphical)

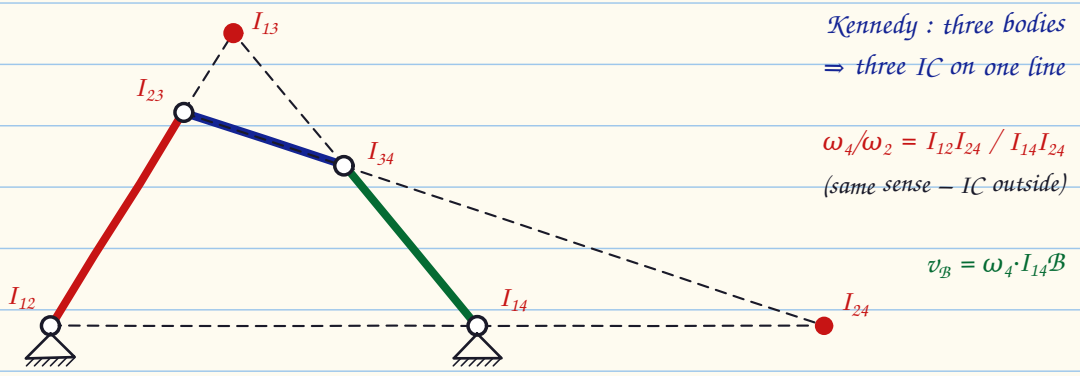
- $a_A = \omega_2^2\, r$, directed A → O (no tangential part for constant ω_2). Draw o′a′. From a′ draw $a^r_{BA} = (ab)^2/AB$ parallel to BA, direction B → A. From its end draw a perpendicular line (tangential component). From o′ draw the line parallel to the slider path – intersection b′ ⇒ a_B = o′b′ . $\alpha_3 = a^t_{BA}/AB$.
- **Klein's construction** : a graphical shortcut on the configuration diagram itself. Extend the connecting rod to cut the perpendicular through O to the line of stroke at M ; then OM represents the piston velocity ($v_B = \omega \cdot OM$) and AM the velocity of B relative to A. A second set of auxiliary circles on OA and AB gives the acceleration vectors without drawing a separate polygon.

3. Instantaneous Centre Method (graphical velocity)

Instantaneous centre (IC) of two bodies in relative plane motion = the point (of both bodies) that has zero relative velocity – at that instant body 2 appears to rotate about it.

- Number of IC : $\mathcal{N} = n(n-1)/2$ (n = number of links). Four-bar → 6 ; slider-crank → 6. Types : **fixed** (I_{12}, I_{14} at frame pins), **permanent** (pin joints between moving links I_{23}, I_{34}), **neither fixed nor permanent** (I_{13}, I_{24}).
- **Kennedy's (Aronhold–Kennedy) theorem** : the three instantaneous centres of three bodies in relative motion lie on a straight line.

Instantaneous centres of a four-bar ($\mathcal{N} = 6$)



Locating the centres (four-bar)

- I_{12}, I_{23}, I_{34}, I_{14} = pin centres. I_{13} = intersection of lines $I_{12}I_{23}$ and $I_{14}I_{34}$. I_{24} = intersection of $I_{12}I_{14}$ (frame line) and $I_{23}I_{34}$ (coupler line).
- Slider-crank : I_{12} = O, I_{23} = A, I_{34} = B, I_{14} = at infinity ⊥ to the slider path ; I_{13} = intersection of OA (extended) and the perpendicular to the slider path through B ; I_{24} at infinity (moves ∥ to the path).
- Rolling contact : the IC is at the point of contact. Sliding contact : the IC lies on the common normal at the point of contact.

Velocity from IC

$$v = \omega \times (\text{distance from IC}) \quad \omega_4/\omega_2 = (I_{12}I_{24}) / (I_{14}I_{24})$$

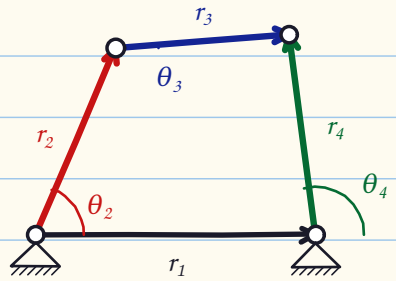
- I_{24} is common to links 2 and 4 so its velocity can be computed from either : $v = \omega_2 \cdot I_{12}I_{24} = \omega_4 \cdot I_{14}I_{24}$. If I_{24} lies between I_{12} and I_{14} the rotation senses are opposite, if outside the same.

Ex 2. (slider-crank above, $\theta = 45°$) : $x_B = r\cos\theta + \sqrt{(l^2 - r^2\sin^2\theta)} = 70.7 + 393.7 = 464.4$ mm ; $I_{13}B = x_B \tan\theta = 464.4$ mm ; $OI_{13} = x_B/\cos\theta = 656.8$ mm $\Rightarrow I_{13}A = 556.7$ mm. $\omega_3 = v_A/I_{13}A = 3.1416/0.5567 =$ 5.64 rad/s ; $v_B = \omega_3 \cdot I_{13}B = 5.64 \times 0.4644 =$ 2.62 m/s.

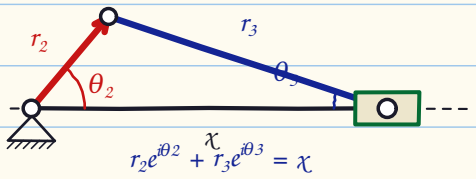
- Advantages : quick for velocity of a point ; disadvantages : not for acceleration ; poor accuracy when centres are far away.

4. Loop-Closure (Vector-Loop) Equations

Four-bar vector loop



Slider-crank vector loop



Treat each link as a vector $r_i e^{i\theta_i}$; the sum of vectors round a closed loop is zero. Differentiate for velocity and acceleration (analytic, exact, computer-ready).

(a) Slider-crank (in-line)

$$r_2\cos\theta_2 + r_3\cos\theta_3 = x \qquad r_2\sin\theta_2 + r_3\sin\theta_3 = 0$$

- θ_3 (angle of the rod vector A→B) is negative : $\sin\theta_3 = -(r_2/r_3)\sin\theta_2$.
- **Velocity** (differentiate w.r.t. time) : $\omega_3 = -r_2\omega_2\cos\theta_2 / (r_3\cos\theta_3)$; $\dot{x} = -r_2\omega_2\sin\theta_2 - r_3\omega_3\sin\theta_3$.
- **Acceleration** : $\alpha_3 = [\, r_2\omega_2^2\sin\theta_2 + r_3\omega_3^2\sin\theta_3 \,] / (r_3\cos\theta_3)$; $\ddot{x} = -r_2\omega_2^2\cos\theta_2 - r_3\alpha_3\sin\theta_3 - r_3\omega_3^2\cos\theta_3$ ($\alpha_2 = 0$).

Ex 3. $r_2 = 0.1$, $r_3 = 0.4$, $\theta_2 = 45°$, $\omega_2 = 31.42$ rad/s : $\sin\theta_3 = -0.1768$ ($\theta_3 = -10.18°$) ; $\omega_3 = -0.1(31.42)(0.7071)/(0.4 \times 0.9843) =$ −5.64 rad/s (clockwise, opposite to crank) ; $\dot{x} = -2.221 - 0.4(-5.64)(-0.1768) =$ −2.62 m/s ; $\alpha_3 = (69.8 - 2.25)/0.3937 = 171.5$ rad/s² ; $\ddot{x} = -69.8 + 12.1 - 12.5 =$ −70.2 m/s² (toward O).

Ex 3(a). Four-bar : $r_1 = 100$, $r_2 = 40$, $r_3 = 90$, $r_4 = 60$ mm ; $\theta_2 = 60°$, $\omega_2 = 10$ rad/s ($\alpha_2 = 0$). Solving the loop : $\theta_3 = 16.1°$, $\theta_4 = 83.8°$. Velocity matrix gives $\omega_3 = -1.94$ rad/s, $\omega_4 = 4.99$ rad/s ⇒ $v_A = 0.40$ m/s, $v_B = r_4\omega_4 = 0.30$ m/s. Acceleration : $\alpha_3 = 27.5$ rad/s², $\alpha_4 = 47.8$ rad/s².

(b) Four-bar linkage

$$r_2 e^{i\theta_2} + r_3 e^{i\theta_3} - r_4 e^{i\theta_4} - r_1 = 0$$

- *Real and imaginary parts give two equations.* ***Position*** *(Freudenstein) with input θ_2, output θ_4 :*

$$K_1 \cos\theta_4 - K_2 \cos\theta_2 + K_3 = \cos(\theta_2 - \theta_4)\quad K_1 = r_1/r_2,\ K_2 = r_1/r_4,\ K_3 = (r_2^2 - r_3^2 + r_4^2 + r_1^2)/(2r_2r_4)$$

- ***Velocity*** *(differentiate) : $r_2\omega_2\, e^{i\theta_2} + r_3\omega_3\, e^{i\theta_3} - r_4\omega_4\, e^{i\theta_4} = 0$ ⇒ in matrix form*

$$[\, -r_3\sin\theta_3 \;\; r_4\sin\theta_4 \;;\; r_3\cos\theta_3 \;\; -r_4\cos\theta_4 \,]\ \{\, \omega_3,\ \omega_4 \,\} = r_2\omega_2\ \{\, \sin\theta_2,\ -\cos\theta_2 \,\}$$

- ***Acceleration*** *: differentiate again – the tangential terms (α_3, α_4) are the unknowns and ω^2 terms are on the right-hand side. Solve with Cramer's rule or numerically (Newton-Raphson).*
- *Advantages : accuracy, complete cycle in one program ; extends to any linkage, synthesis and optimisation.*

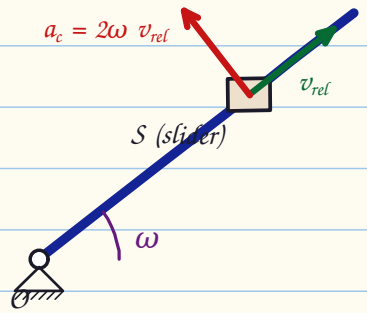
Procedure for a general mechanism

- *(1) Choose independent loops (number = $j - n + 1$). (2) Write vector-loop equations with angles measured from a fixed axis. (3) Solve position for unknown angles (closed-form or iterative). (4) Differentiate once – linear in velocities. (5) Differentiate twice – linear in accelerations.*

5. Coincident Points & Coriolis Acceleration

Coincident points *: two points of different links that occupy the same position at an instant (e.g. a slider on a rotating link and the point of the link under it).*

Coriolis component



Slider on a rotating link

$a_S = a_P + a_{S/P}^{rel} + 2\ \omega \times v_{rel}$

P = coincident point on link

Direction : rotate v_{rel} through 90° in the direction of ω

Exists when : $v_{rel} \neq 0$ and $\omega \neq 0$

Let P be the point on the rotating link 2 (angular velocity ω) and S the slider on it (relative velocity $v_{S/P} = v_{rel}$ along the link). Then

$$v_S = v_P + v_{rel}\ a_S = a_P + a_{rel} + a_c$$

$$a_c\ (Coriolis) = 2\ \omega\ v_{rel}$$

- Components of a_S : (1) a_P (radial $\omega^2 r$ and tangential αr of the link point), (2) a_{rel} = relative acceleration along the slot (sliding), (3) Coriolis $a_c = 2\omega v_{rel}$, perpendicular to the slot, direction obtained by rotating v_{rel} through 90° in the direction of ω.
- Origin : the sliding velocity changes direction due to rotation of the link, while the link's point velocity changes magnitude because the slider moves to a larger radius – two equal contributions (ωv_{rel} each).
- It exists only when the slider/point has a relative velocity on a rotating link ($v_{rel} \neq 0$, $\omega \neq 0$). Zero when $\omega = 0$ (translation) or $v_{rel} = 0$ (rigid) – so none in the simple slider-crank (slider moves in a fixed guide).
- Examples : crank & slotted lever, Whitworth quick return, oscillating-cylinder engine, Geneva, cam with sliding roller, aircraft in a turn, Foucault's pendulum, cyclones, ocean currents.

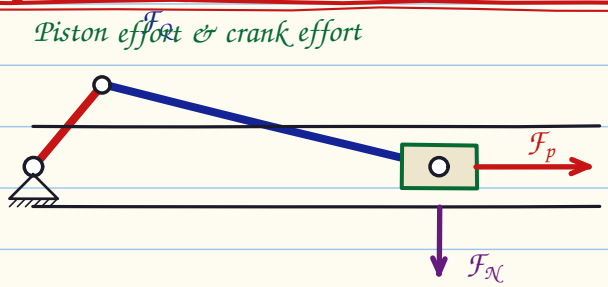
Ex 4. A slider moves outward in a radial slot of a link rotating at $\omega = 5$ rad/s (constant) ; at radius 0.3 m, $v_{rel} = 0.4$ m/s, $a_{rel} = 0$. Absolute speed : $v = \sqrt{((0.3\times5)^2 + 0.4^2)} = 1.55$ m/s. Acceleration : radial = $\omega^2 r = 7.5$ m/s² (toward centre) ; Coriolis = $2(5)(0.4) = 4$ m/s² ⊥ slot ⇒ $a = \sqrt{(7.5^2 + 4^2)} = 8.5$ m/s².

Procedure for a quick-return (crank & slotted lever)

- Crank pin A (on link 2) and the slotted-lever point A' (coincident, on link 3 = lever) : $v_A = v_{A'} + v_{A/A'}$ (sliding along lever) ⇒ gives lever angular velocity $\omega_3 = v_{A'}/(PA)$.
- Acceleration : a_A (radial, crank) $= a_{A'}^{\,r} + a_{A'}^{\,t} + a_c + a_{rel}$. Solve the polygon for α_3 and a_{rel}.

6. Dynamics of the Slider-Crank Mechanism

Piston effort & crank effort



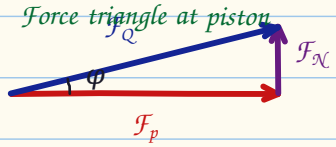
$\sin\varphi = \sin\theta / n$, $n = l/r$

$F_N = F_p \tan\varphi$

$F_Q = F_p / \cos\varphi$

$T = F_p\, r \sin(\theta+\varphi)/\cos\varphi$

Force triangle at piston



Forces on the piston and crank

- Piston (gas) force $F = (\pi/4)D^2\, p$ (net of friction). Inertia of reciprocating parts (mass m) :

$$F_I = m\, r\, \omega^2\, (\cos\theta + \cos 2\theta/n) \text{ primary} = mr\omega^2\cos\theta \text{ , secondary} = mr\omega^2\cos 2\theta/n$$

- **Net (piston) effort** $F_p = F - F_I - mg$ (vertical engines add weight ; sign for inertia opposes acceleration).
- Thrust on the cylinder wall $F_N = F_p \tan\varphi$; thrust along the connecting rod $F_Q = F_p/\cos\varphi$; force on the crank pin = F_Q.
- **Crank effort (turning moment)** :

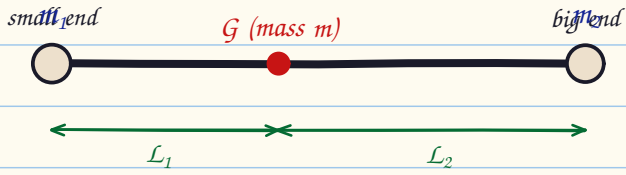
$$T = F_Q\, r \sin(\theta+\varphi) = F_p\, r \sin(\theta+\varphi)/\cos\varphi \approx F_p\, r\, (\sin\theta + \sin 2\theta/(2\sqrt{(n^2 - \sin^2\theta)}))$$

- Radial component $F_R = F_Q \cos(\theta+\varphi)$ loads the crankshaft bearings ; reaction torque on the frame = T (equal & opposite) ⇒ tends to overturn the engine : T = F_N × (crank-shaft height).
- Mean torque from the indicator diagram ; inertia causes torque fluctuations (flywheel – Unit V) and shaking force F_I on the frame (balancing – Unit IV).

Ex 5. $m = 20$ kg reciprocating mass, $r = 0.1$ m, $N = 300$ rpm, $n = 4$, $\theta = 45°$: $F_I = 20(0.1)(31.42)^2(0.7071 + 0) =$ **1396 N** (acts opposite to piston acceleration). If gas force = 12 kN : $F_p = 12000 - 1396 = 10604$ N ; $\sin\varphi = 0.1768 \Rightarrow \varphi = 10.2°$; $F_N = 10604 \tan 10.2° =$ **1908 N** ; $T = 10604 \times 0.1 \times \sin(55.2°)/\cos 10.2° =$ **883 N·m**.

Dynamically equivalent system of connecting rod

Dynamically equivalent system of the connecting rod



1) $m_1 + m_2 = m$

2) $m_1L_1 = m_2L_2$

3) $m_1L_1^2 + m_2L_2^2 = I_G$

($K^2 = L_1L_2$)

The distributed mass of the connecting rod is replaced by two point masses (at small end & big end) so that inertia effects are equal :

- Three conditions : (1) total mass equal $m_1 + m_2 = m$; (2) CG unchanged $m_1L_1 = m_2L_2$; (3) moment of inertia equal $m_1L_1^2 + m_2L_2^2 = mk_G^2$.
- Condition 3 requires $L_1L_2 = k_G^2$. If only two masses are placed at the ends ($L_1 + L_2 = l$) condition 3 is not satisfied – a correction **torque** $T_c = m(L_1L_2 - K^2)\alpha$ must be added. Usual : 1/3 mass at crank pin, 2/3 at piston pin (approx.).
- Inertia of the equivalent system : m_2 (small end) adds to the reciprocating mass ; m_1 (big end) is a rotating mass at the crank-pin radius.

8. Special Topics in Velocity Analysis

Velocity image & velocity of a point on a link

- For a point C on link AB, in the velocity polygon the line ab is the "image" of AB : point c divides ab in the same ratio as C divides AB (ac/ab = AC/AB). For a link with three points the triangle abc is similar to ABC and rotated 90° (**velocity image principle**).
- Velocity of a point on a link given its two other points : draw the image first, join the pole o to c ⇒ v_C. Needed for centre of gravity velocity in dynamics.

Velocity of rubbing at a pin joint

- Rubbing velocity at a pin of radius r between two links : $(\omega_1 \pm \omega_2)\, r$ — '+' when the links rotate in opposite directions, '−' when in the same direction. Used to find the work lost in friction at joints and wear.
- At a revolute pair of a slider-crank : crank pin $(\omega_2 - \omega_3) r_p$; crosshead pin $\omega_3\, r_c$; crank shaft $\omega_2\, r_s$.

Klein's construction (slider-crank)

- Extend the connecting rod AB to meet the perpendicular through O to the line of stroke at M. Triangle OAM is similar to the velocity triangle $\Rightarrow v_B \propto OM$, $v_{BA} \propto AM$, and (for constant ω) a_B is obtained from the intersections of auxiliary circles : draw a circle on AB as diameter and a circle with centre A, radius AB^2/OA..., the quadrilateral gives $a_B = \omega^2 \times$ (length). Saves the polygon.

Mechanisms with sliders on rotating links — using IC

- Crank & slotted lever : I_{13} (crank-lever) lies on the line of the crank pin and the lever pivot ; the velocity of the sliding block relative to the slot is found by resolving v_A along and perpendicular to the lever.
- Scotch yoke : $x = r\sin\theta \Rightarrow v = r\omega\cos\theta$, $a = -r\omega^2\sin\theta$; the IC of the yoke is at infinity (pure translation).

9. Additional Solved Problems

P1 (IC ratio). Four-bar with $I_{12}I_{24} = 0.20$ m and $I_{14}I_{24} = 0.08$ m (I_{24} outside the frame pins $\Rightarrow$ same sense), $\omega_2 = 20$ rad/s : $v_{I24} = 20 \times 0.20 = 4$ m/s $\Rightarrow \omega_4 = 4/0.08 =$ **50 rad/s** ($\omega_4/\omega_2 = 2.5$).

P2 (relative velocity). A rod $AB = 0.5$ m has $v_A = 3$ m/s along the rod's perpendicular-free direction ; if $v_B = 5$ m/s and $v_A \perp v_B$: $v_{BA} = \sqrt{(5^2 - 3^2)} = 4$ m/s $\perp AB \Rightarrow \omega_{AB} = 4/0.5 =$ **8 rad/s**.

P3 (piston speed). Engine $r = 75$ mm, $l = 300$ mm, 1200 rpm : $\omega = 125.7$ rad/s ; at $\theta = 90°$ $v = r\omega(1 + 0) = 9.42$ m/s ($r\omega\sin\theta + r\omega\sin 180°/2n = 9.42$) ; mean piston speed $= 2\,L\,N/60 = 2(0.15)(1200)/60 =$ **6.0 m/s**.

P4 (max acceleration). Same engine : a_{max} (IDC) $= r\omega^2(1 + 1/n) = 0.075(125.7)^2(1 + 0.25) =$ **1481 m/s²** ; at ODC $r\omega^2(1 - 1/n) = 889$ m/s².

P5 (Coriolis in QR). Lever rotating at $\omega = 4$ rad/s, slider moving along it at $v_{rel} = 0.6$ m/s : $a_c = 2 \times 4 \times 0.6 = 4.8$ m/s².

Common mistakes

- Direction of a^r_{BA} is $B \to A$ (towards the reference point) ; tangential is $\perp AB$, sense consistent with α.
- Forgetting the Coriolis term for a slider on a rotating link or using ω of the wrong link (use ω of the link carrying the slot).
- Mixing rpm and rad/s ; using degrees in calculator mode for sin/cos of radians.

10. Quick Formula Sheet – Unit II

Item	Formula
Relative motion	$v_{BA} = \omega \cdot AB$; $a^r_{BA} = \omega^2 AB$; $a^t_{BA} = \alpha \cdot AB$
Piston	$v = r\omega(\sin\theta + \sin 2\theta/2n)$; $a = r\omega^2(\cos\theta + \cos 2\theta/n)$
Rod angular velocity	$\omega_3 = \omega \cos\theta / \sqrt{(n^2 - \sin^2\theta)}$
Instantaneous centres	$N = n(n-1)/2$; $\omega_4/\omega_2 = I_{12}I_{24}/I_{14}I_{24}$
Loop closure	$\Sigma\, r_i e^{i\theta_i} = 0$; differentiate for v, a
Coriolis	$a_c = 2\omega\, v_{rel}$
Inertia force	$F_I = mr\omega^2(\cos\theta + \cos 2\theta/n)$
Forces	$F_N = F_p \tan\varphi$; $F_Q = F_p/\cos\varphi$; $T = F_p\, r \sin(\theta+\varphi)/\cos\varphi$

11. Important Questions

- Explain the relative-velocity method; draw velocity and acceleration polygons for a slider-crank mechanism and derive piston velocity & acceleration.
- State Kennedy's theorem; locate all instantaneous centres of a four-bar and slider-crank mechanism ; find ω_4.
- Write the loop-closure equations for a four-bar and a slider-crank and obtain the velocity and acceleration equations.
- What are coincident points? Derive the Coriolis component of acceleration and state its direction rule.
- Obtain the piston effort, thrust on the cylinder wall, force along the connecting rod and crank effort of a slider-crank.

- Explain the dynamically equivalent two-mass system of a connecting rod.